t3-w09 — condition C2

Need for statistic — Need for statistics: purposes, and the collection and tabulation of data

> C1 plus the lesson note as source evidence.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	**present**
Materials	what the lesson was represented with	prompt	**present**
Position	where this objective sat in the sequence	prompt	**present**
Method	the shape of the teaching, as event labels	prompt	**present**
Enacted content	what was actually explained and worked through	prompt	**present**
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	**present**
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteGroundedSection`, declaring: `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Need for statistic

`subtopic`

Need for statistics: purposes, and the collection and tabulation of data

`term`

Third Term

`week`

9

`lesson_objectives`

• explain the purpose of statistic
• analyse and interpret statistical data presented in tables

entry_behaviour

• Students are familiar with counting and recording information.

instructional_materials

• classroom register, bottle tops, pencils or counters.

Objective 1 — explain the purpose of statistic

objective_text

explain the purpose of statistic

current_step

1

instructional_events

• present_content

assessment_points

• (a) Define statistic

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "statistics is the science of collecting, organizing, presenting, analyzing and
interpreting\ndata.\n\n#### Meaning of Data\n\nData are facts or information collected for a
particular purpose.\n\n#### Importance of statistics\n\nIt helps us to\n\n- (a) Collect
Information\n- (b) organize Information\n- (c) Compare Information\n- (d) Make decisions\n- (e)
Present Information Clearly."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — analyse and interpret statistical data presented in tables

`objective_text`

analyse and interpret statistical data presented in tables

`current_step`

2

`assessment_points`

• (b) State three methods of collecting data.
• (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Collection of Data\n\nData can be collected through\n\n1. observation\n2.
counting\n3. Interview\n4. Questionnaire.\n\n#### Tally Marks\n\nTally Marks are used to record
data as they are collected.\n\n**Example:** The fig below show the same data collected in two
different ways by two student\n\n(a) bus, car, car, car, lorry, bicycle, bicycle, car, car,
lorry, bicycle, car.\n\nThe above data can be represented in Tally\n\n| Vehicles | Tally | Total
|\n|---|---|---|\n| car | \\|\\|\\|\\|/ \\| | 6 |\n| bus | \\| | 1 |\n| lorry | \\|\\| | 2 |\n|
bicycle | \\|\\|\\| | 3 |\n| Motor Bike | | |\n\n#### Frequency Table\n\nA frequency table is a
table showing how many times each item occurs in a set of data.\n\n**Examples:**\n\nFavourite
colours of student:\n\n| colour | frequency |\n|---|---|\n| Red | 18 |\n| Blue | 20 |\n| Green |
14 |\n| Yellow | 16 |\n| White | 15 |\n| Pink | 30 |"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `instructional_events`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Why We Use Statistics

> EXPLANATION
> Statistics is the science of collecting, organising, presenting, analysing and interpreting data.
>
> **Data** are facts or information gathered for a particular purpose.
>
> **Why statistics matters**
> - **Collect information** – it helps us gather the facts we need.
> - **Organise information** – it puts data in a clear order so we can see patterns.
> - **Compare information** – it lets us see how different sets of data relate to each other.
> - **Make decisions** – it gives a solid basis for choosing what to do.
> - **Present information clearly** – it turns numbers into tables, charts or summaries that are easy to understand.

> WORKED_EXAMPLE
> **Worked example**
> Imagine you want to know which colour of bottle tops is most popular in your class. You ask each student to bring a bottle top and record its colour.
>
> 1. **Collect** the colours (data).
> 2. **Organise** them into a table showing how many of each colour you have.
> 3. **Compare** the totals to see which colour appears most often.
> 4. **Decide** whether to order more of that colour for a school project.
> 5. **Present** the result in a simple table so everyone can see the answer at a glance.
>
> This process shows all five purposes of statistics in action.

1. 1. List two reasons why collecting data is important in everyday life.
answer: It provides the facts needed to understand a situation and helps us make informed decisions.

2. 2. You have recorded the number of books read by each student in a class: 3, 5, 2, 4, 5. Explain how you could use statistics to decide which reading level to recommend for the next term.
answer: By organising the numbers into a table or chart, you can see the typical range (2-5 books). Comparing the data shows most students read 4-5 books, so a reading level that matches that range would be appropriate.

3. 3. Define the word *statistics* in your own words.
answer: Statistics is the study of how to collect, organise, present, analyse and interpret data.

Objective 2

Interpreting Data in Tables

> EXPLANATION
> Statistics helps us understand the world by collecting information and organising it so we can see patterns.
>
> **Ways to collect data**
> 1. Observation – watching what happens and noting it down.
> 2. Counting – tallying items or events.
> 3. Interview – asking people questions directly.
> 4. Questionnaire – giving people a set of written questions.
>
> **Tally marks** are a quick way to record data as you collect it. Each group of four vertical lines \|\|\|\| is crossed with a diagonal /, making it easy to count groups of five.
>
> **Frequency tables** show how many times each value occurs. They are useful for spotting the most common items and for comparing quantities.

> WORKED_EXAMPLE
> ### Example: Making a frequency table
>
> Suppose we record the number of bottle tops each student collects: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.
>
> 1. **Tally the data**
> | Value | Tally | Total |
> |---|---|---|
> | 1 | \|\|\|\| \| | 3 |
> | 2 | \|\|\|\| \| | 3 |
> | 3 | \|\|\|\|\| \| | 4 |
> | 4 | \|\|\|\| \| | 3 |
>
> 2. **Convert tallies to a frequency table**
> | Value | Frequency |
> |---|---|
> | 1 | 3 |
> | 2 | 3 |
> | 3 | 4 |
> | 4 | 3 |
>
> The table tells us that the number 3 was collected most often.

1. (b) List three methods you could use to collect data for a school survey.
answer: Observation, counting, interview (any three of the four listed methods).

2. (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.
answer: Value | Frequency 1 | 3 2 | 3 3 | 4 4 | 3

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	2	Scope	**grounded**
Explanation the principle, rule or relationship to review	2	Enacted content	**grounded**
	2	Enacted content	**grounded**

Worked example a problem with its solution worked through			
Figure a drawing the learner reads from	0	nothing in the input	not produced
Practice what the learner does	5	Assessment written from the teacher's evaluation questions instead of her set work	**substituted**
Answer what the learner checks against	5	Enacted content	**grounded**